

Basics of Markov Chain Simulation:

Gibbs Sampler, Metropolis–Hastings, Convergence and Effective Sample Size

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Based on *Bayesian Data Analysis* by Gelman et al. [1], Chapter 11. An R example with trace plots, $\hat{R}$ and effective sample size (ESS) is included. Code available on request.

1 Markov Chain Simulation

When direct simulation (Chapter 10) is impractical, especially in high dimensions, we build a **Markov chain** $\theta^1, \theta^2, \dots$ whose distribution converges to the target $p(\theta | y)$ [1, Ch. 11].

- A sequence $\theta^1, \theta^2, \dots$ is a **Markov chain** if θ^t depends on the past only through θ^{t-1} , via a **transition distribution** $T_t(\theta^t | \theta^{t-1})$.
- The chain is constructed so that its **stationary distribution** is the posterior $p(\theta | y)$.
- Draws from early iterations depend on the starting point, so they are discarded as **warm-up** (burn-in). Later draws are approximately from $p(\theta | y)$ but are *dependent*.
- Because draws are correlated, the number of draws needed is larger than for independent simulation; this is measured by the effective sample size (Section 5).

Transition vs. proposal (jumping) distribution.

The transition distribution is a property of the Markov chain. In the Metropolis algorithm it is a mixture of the proposal (jumping) distribution J_t and a point mass at the current value, since a rejected proposal keeps the chain where it is.

2 Gibbs Sampler

Partition $\theta = (\theta_1, \dots, \theta_d)$. The Gibbs sampler draws each component in turn from its full conditional distribution given the current values of all the others [1, Sec. 11.1].

Algorithm 1 Gibbs sampler

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1: Choose a starting point  $\theta^0$ .
2: for  $t = 1, 2, \dots$  do
3:   for  $j = 1, \dots, d$  do
4:     Draw  $\theta_j^t \sim p(\theta_j \mid \theta_1^t, \dots, \theta_{j-1}^t, \theta_{j+1}^{t-1}, \dots, \theta_d^{t-1}, y)$ .
5:   end for
6: end for

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Key points.

- It is useful when the full conditionals are easy to sample from (for example, conjugate or semi-conjugate models).
- It is a special case of Metropolis–Hastings in which every proposal is accepted.
- Convergence is slow when components are strongly correlated in the posterior, because each step moves along a coordinate axis.

Example 2.1 (Bivariate normal). For $(\theta_1, \theta_2) \sim N\left(0, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$ the full conditionals are

$$\theta_1 \mid \theta_2 \sim N(\rho \theta_2, 1 - \rho^2), \quad \theta_2 \mid \theta_1 \sim N(\rho \theta_1, 1 - \rho^2).$$

This is the target used in Section 6 with $\rho = 0.8$.

3 Metropolis and Metropolis–Hastings

3.1 Metropolis algorithm

Use a **symmetric** jumping distribution, $J_t(\theta_a \mid \theta_b) = J_t(\theta_b \mid \theta_a)$ [1, Sec. 11.2].

Algorithm 2 Metropolis algorithm

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1: Choose a starting point  $\theta^0$  with  $p(\theta^0 \mid y) > 0$ .
2: for  $t = 1, 2, \dots$  do
3:   Draw a proposal  $\theta^* \sim J_t(\theta^* \mid \theta^{t-1})$ .
4:   Compute the ratio  $r = \frac{p(\theta^* \mid y)}{p(\theta^{t-1} \mid y)}$ .
5:   Set  $\theta^t = \theta^*$  with probability  $\min(r, 1)$ , and  $\theta^t = \theta^{t-1}$  otherwise.
6: end for

```

Only the *ratio* of posterior densities is needed, so the unnormalized density $q(\theta \mid y) = p(y \mid \theta)p(\theta)$ suffices.

3.2 Metropolis–Hastings

For a non-symmetric jumping distribution, correct the ratio by the *ratio of ratios*:

$$r = \frac{p(\theta^* | y) / J_t(\theta^* | \theta^{t-1})}{p(\theta^{t-1} | y) / J_t(\theta^{t-1} | \theta^*)}. \quad (1)$$

The acceptance rule is unchanged: accept with probability $\min(r, 1)$.

3.3 Why the posterior is stationary

Take two points θ_a, θ_b , labelled so that $p(\theta_b | y) J_t(\theta_a | \theta_b) \geq p(\theta_a | y) J_t(\theta_b | \theta_a)$. If $\theta^{t-1} \sim p(\cdot | y)$, the joint density of a move $\theta_a \rightarrow \theta_b$ is

$$p(\theta_a | y) J_t(\theta_b | \theta_a),$$

since the acceptance probability is 1 in this direction. The joint density of the reverse move $\theta_b \rightarrow \theta_a$ is

$$p(\theta_b | y) J_t(\theta_a | \theta_b) \cdot \frac{p(\theta_a | y) / J_t(\theta_a | \theta_b)}{p(\theta_b | y) / J_t(\theta_b | \theta_a)} = p(\theta_a | y) J_t(\theta_b | \theta_a),$$

which is the same. The joint distribution of (θ^{t-1}, θ^t) is therefore symmetric, so θ^t and θ^{t-1} have the same marginal distribution and $p(\theta | y)$ is stationary. The stationary distribution is unique if the chain is irreducible, aperiodic and not transient.

Tuning the jumping scale.

- A jumping scale that is too small gives a high acceptance rate but tiny steps, so the chain explores slowly and is highly autocorrelated.
- A scale that is too large gives many rejected proposals, so the chain gets stuck at the same value.
- The best efficiency is in between; the optimal jumping scale and acceptance rate for normal targets are discussed in Chapter 12 of [1].

4 Inference and Assessing Convergence

4.1 Practical recipe

- Simulate $m \geq 2$ independent chains from **overdispersed starting points**.
- Discard the first half of each chain as warm-up.
- **Split** each remaining chain in half, giving $2m$ sequences of length n , so that lack of stationarity within a chain is also detected.
- Compare between-sequence and within-sequence variation [1, Sec. 11.4].

4.2 Potential scale reduction factor $\hat{R}$

For a scalar estimand ψ with simulations ψ_{ij} ($i = 1, \dots, n$; $j = 1, \dots, m$, after splitting):

$$B = \frac{n}{m-1} \sum_{j=1}^m (\bar{\psi}_{\cdot j} - \bar{\psi}_{\cdot \cdot})^2, \quad W = \frac{1}{m} \sum_{j=1}^m s_j^2, \quad s_j^2 = \frac{1}{n-1} \sum_{i=1}^n (\psi_{ij} - \bar{\psi}_{\cdot j})^2, \quad (2)$$

$$\widehat{\text{Var}}^+(\psi | y) = \frac{n-1}{n} W + \frac{1}{n} B, \quad \hat{R} = \sqrt{\frac{\widehat{\text{Var}}^+(\psi | y)}{W}}. \quad (3)$$

- $\hat{R} \rightarrow 1$ as $n \rightarrow \infty$. Large values mean the chains have not mixed, so more iterations are needed.
- BDA3 recommends continuing until $\hat{R} < 1.1$ for all scalar estimands of interest; later work recommends the stricter threshold $\hat{R} < 1.01$ [3].
- $\hat{R}$ can be slightly below 1 by chance.

5 Effective Number of Simulation Draws

With $N = mn$ dependent draws, the variance of the sample mean is larger than $\text{Var}(\psi)/N$. The **effective sample size** is the number of independent draws that would give the same precision [1, Sec. 11.5].

The **variogram** at lag t is

$$V_t = \frac{1}{m(n-t)} \sum_{j=1}^m \sum_{i=t+1}^n (\psi_{ij} - \psi_{i-t,j})^2, \quad (4)$$

and the estimated autocorrelation at lag t is

$$\hat{\rho}_t = 1 - \frac{V_t}{2 \widehat{\text{Var}}^+}. \quad (5)$$

The effective sample size is

$$\hat{n}_{\text{eff}} = \frac{mn}{1 + 2 \sum_{t=1}^T \hat{\rho}_t}, \quad (6)$$

where T is the first odd positive integer for which $\hat{\rho}_{T+1} + \hat{\rho}_{T+2}$ is negative; summing stops there to avoid adding noise from large lags.

Reading ESS.

- $\hat{n}_{\text{eff}} \approx mn$ means the draws are nearly independent. $\hat{n}_{\text{eff}} \ll mn$ means strong serial correlation.
- The Monte Carlo standard error of a posterior mean is about $\widehat{\text{sd}}(\psi | y) / \sqrt{\hat{n}_{\text{eff}}}$.
- ESS is computed per scalar quantity, so check it for each estimand of interest.

6 R Example: Gibbs and Metropolis on a Bivariate Normal

Target: $(\theta_1, \theta_2) \sim N\left(0, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$ with $\rho = 0.8$.

Setup. For each sampler, $m = 4$ chains of 2000 iterations were run from the overdispersed starting points $(\pm 4, \pm 4)$. The first 1000 iterations of each chain are warm-up (shaded in the trace plots) and are discarded, leaving 4000 draws per sampler. Three samplers are compared:

- (1) the Gibbs sampler of Example 2.1;
- (2) random-walk Metropolis with jumping distribution $N(\theta^{t-1}, 0.2^2 I)$;
- (3) random-walk Metropolis with jumping distribution $N(\theta^{t-1}, 1.5^2 I)$.

$\hat{R}$ and $\hat{n}_{\text{eff}}$ are computed with the formulas of Sections 4–5 on the split post-warm-up chains (implemented in base R in the script; no packages are needed).

6.1 Trace plots

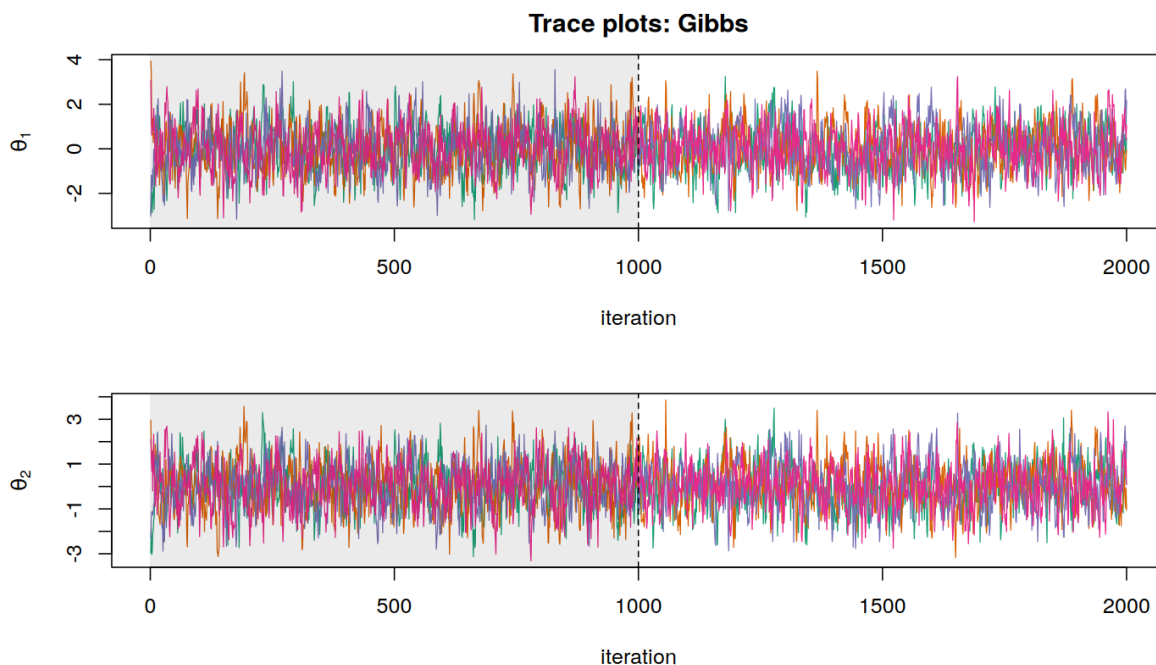


Figure 1: Gibbs sampler: four chains, warm-up shaded. The chains forget their starting points almost immediately.

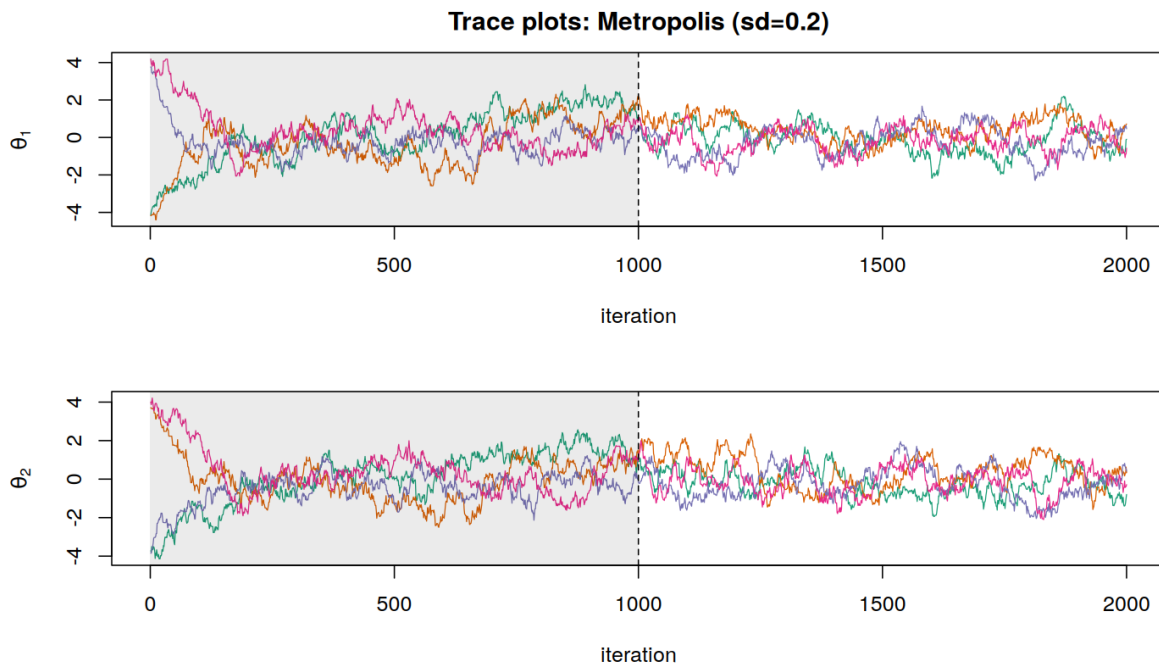


Figure 2: Metropolis with jumping scale 0.2: small steps, slow exploration, and visible wandering even after warm-up.

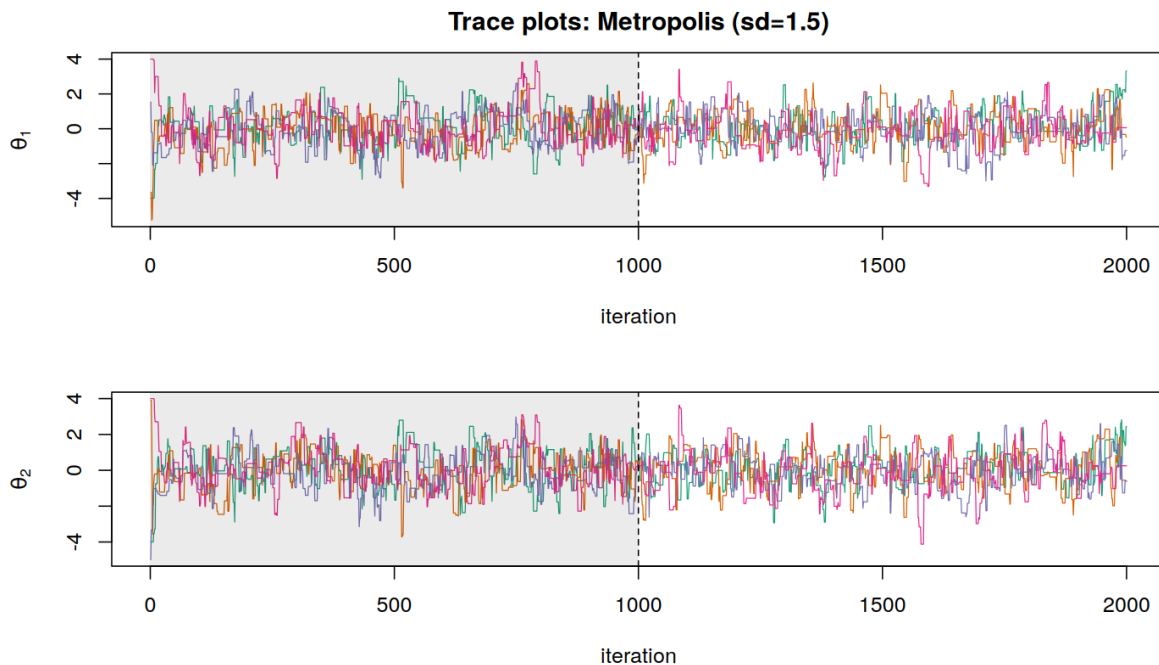


Figure 3: Metropolis with jumping scale 1.5: larger steps, better mixing.

6.2 Draws and autocorrelation

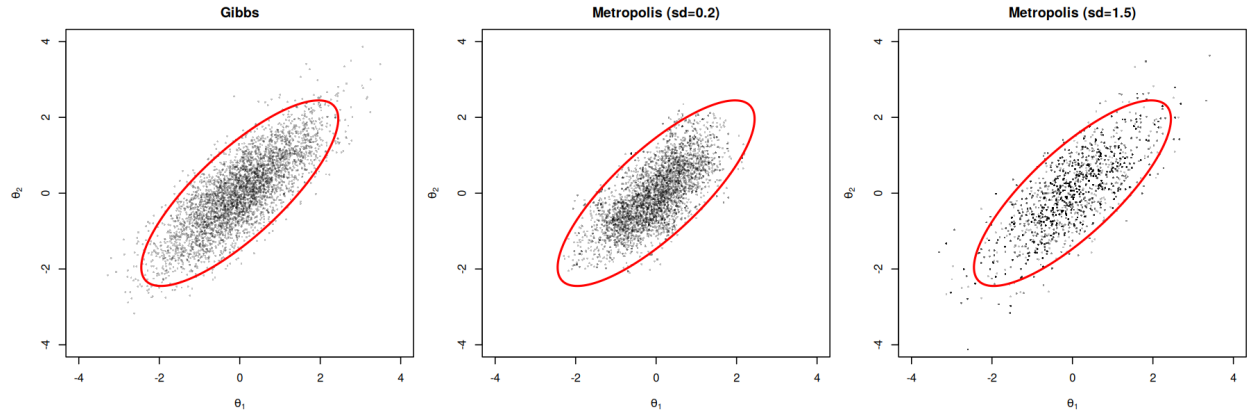


Figure 4: Post-warm-up draws of (θ_1, θ_2) from each sampler with the true 95% probability ellipse in red.

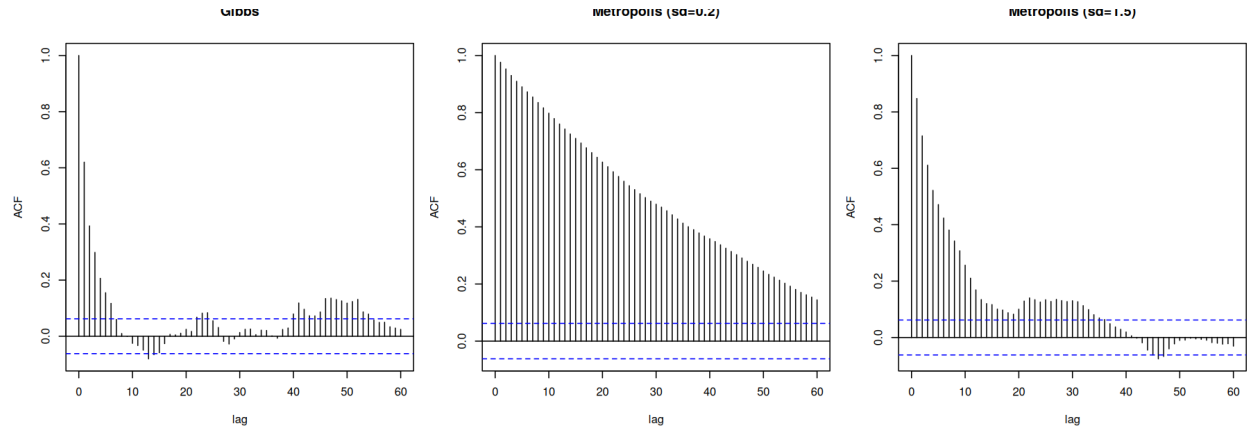


Figure 5: Autocorrelation of θ_1 (first chain, post-warm-up). The Gibbs sampler decays fastest; Metropolis with scale 0.2 decays slowly.

6.3 Convergence and effective sample size

Sampler	Parameter	Mean	SD	$\hat{R}$	ESS	Accept. rate
Gibbs	θ_1	0.02	0.99	1.002	929	—
Gibbs	θ_2	0.02	0.98	1.001	948	—
Metropolis (sd=0.2)	θ_1	-0.03	0.78	1.114	37	0.84
Metropolis (sd=0.2)	θ_2	-0.04	0.81	1.080	37	0.84
Metropolis (sd=1.5)	θ_1	-0.02	0.97	1.010	325	0.27
Metropolis (sd=1.5)	θ_2	0.00	1.00	1.008	292	0.27

Table 1: Posterior mean, standard deviation, split $\hat{R}$, effective sample size $\hat{n}_{\text{eff}}$ (out of 4000 post-warm-up draws) and Metropolis acceptance rate. The true mean is 0 and the true standard deviation is 1 for both coordinates.

Reading the results.

- **Gibbs:** $\hat{R} \approx 1.00$ and $\hat{n}_{\text{eff}} \approx 930\text{--}950$, about 23% of the 4000 draws. As a check, the lag- t autocorrelation of θ_1 under this Gibbs sampler is ρ^{2t} , so the theoretical ratio is $\hat{n}_{\text{eff}}/N = (1 - \rho^2)/(1 + \rho^2) \approx 0.22$, which matches.
- **Metropolis, scale 0.2:** the acceptance rate is high (≈ 0.84) but the steps are tiny. $\hat{R}$ is about 1.1, so the chains have not converged to the BDA3 threshold, $\hat{n}_{\text{eff}}$ is only about 37, and the estimated standard deviations (≈ 0.8) are noticeably below the true value 1.
- **Metropolis, scale 1.5:** acceptance rate ≈ 0.27 , $\hat{R} \approx 1.01$ and $\hat{n}_{\text{eff}}$ about 290–325, roughly $8\times$ more efficient than scale 0.2.

Takeaway.

A high acceptance rate is not the same as good mixing. The raw number of draws (4000 for each sampler) is the same in every row of Table 1, but the information they carry differs by a factor of about 25. Always report $\hat{n}_{\text{eff}}$, not just the number of iterations.

7 Exercise 6: Effective Sample Size

7.1 (a) Variance of the average of correlated simulations

Setup. Let $\psi_1, \psi_2, \dots$ be a stationary sequence (one chain at stationarity) with mean $\mathbb{E}(\psi \mid y)$, variance $\sigma^2 = \text{Var}(\psi \mid y)$ and autocorrelations $\rho_t = \text{corr}(\psi_i, \psi_{i+t})$. Assume $\sum_{t=1}^{\infty} |\rho_t| < \infty$.

One chain. With $\bar{\psi} = \frac{1}{n} \sum_{i=1}^n \psi_i$,

$$\text{Var}(\bar{\psi}) = \frac{1}{n^2} \sum_{i=1}^n \sum_{k=1}^n \text{Cov}(\psi_i, \psi_k) = \frac{\sigma^2}{n^2} \sum_{i=1}^n \sum_{k=1}^n \rho_{|i-k|}. \quad (7)$$

A lag $t \in \{1, \dots, n-1\}$ appears $2(n-t)$ times in the double sum (and lag 0 appears n times), so

$$n \text{Var}(\bar{\psi}) = \sigma^2 \left[1 + 2 \sum_{t=1}^{n-1} \left(1 - \frac{t}{n} \right) \rho_t \right]. \quad (8)$$

Limit. The terms $(1 - t/n)\rho_t \mathbf{1}\{t < n\}$ converge to ρ_t for each fixed t as $n \rightarrow \infty$ and are bounded in absolute value by $|\rho_t|$, which is summable. By dominated convergence,

$$\lim_{n \rightarrow \infty} n \text{Var}(\bar{\psi}) = \sigma^2 \left(1 + 2 \sum_{t=1}^{\infty} \rho_t \right). \quad (9)$$

m independent chains. The chain means $\bar{\psi}_{\cdot j}$ are independent with identical variance, so the grand mean $\bar{\psi}_{\cdot\cdot} = \frac{1}{m} \sum_j \bar{\psi}_{\cdot j}$ has

$$\lim_{n \rightarrow \infty} mn \text{Var}(\bar{\psi}_{\cdot\cdot}) = \left(1 + 2 \sum_{t=1}^{\infty} \rho_t \right) \text{Var}(\psi \mid y), \quad \text{that is,} \quad \text{Var}(\bar{\psi}_{\cdot\cdot}) \approx \frac{\text{Var}(\psi \mid y)}{mn/(1 + 2 \sum_t \rho_t)}. \quad (10)$$

This is the asymptotic formula (11.5) of the exercise. Independent draws would give variance $\text{Var}(\psi | y)/n_{\text{eff}}$, so

$$n_{\text{eff}} = \frac{mn}{1 + 2 \sum_{t=1}^{\infty} \rho_t}, \quad (11)$$

which is estimated in practice by the truncated, variogram-based formula (11.8) of Section 5.

Conditions used.

The derivation needs stationarity (the chain started from the target, or after warm-up), finite variance, and $\sum_t |\rho_t| < \infty$. If the autocorrelations are not summable, n_{eff} is not of order mn and the $1/\sqrt{mn}$ rate of the Monte Carlo error fails.

7.2 (b) $\hat{n}_{\text{eff}}$ as a function of the number of iterations

Example. The bivariate normal with $\rho = 0.8$ from Section 6, sampled with the Gibbs sampler and two Metropolis jumping scales, $m = 4$ chains per sampler. For each run length $n = 200, 400, \dots, 10000$, the first half of the first n iterations is discarded, and $\hat{n}_{\text{eff}}$ of θ_1 is computed from (11.8) on the split chains.

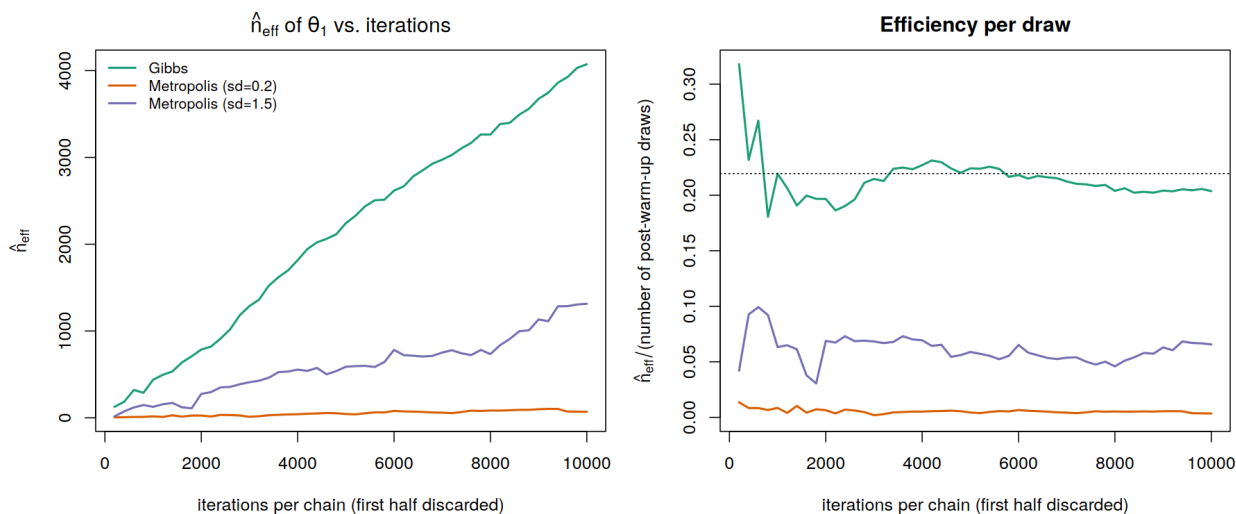


Figure 6: Left: $\hat{n}_{\text{eff}}$ versus run length. Right: $\hat{n}_{\text{eff}}$ divided by the number of post-warm-up draws; the dotted line is the theoretical Gibbs value $(1 - \rho^2)/(1 + \rho^2) \approx 0.22$.

Iterations n	Gibbs	Metropolis (sd=0.2)	Metropolis (sd=1.5)
400	185	7	74
1000	439	17	126
2000	786	26	275
5000	2242	44	589
10000	4073	70	1314

Table 2: $\hat{n}_{\text{eff}}$ of θ_1 after n iterations per chain (4 chains, first half discarded).

Answer.

- $\hat{n}_{\text{eff}}$ is *not* constant, and it *does* grow with the number of iterations, roughly linearly, as one would hope: the number of effectively independent draws is proportional to the number of draws once the chain is stationary.
- For the Gibbs sampler the ratio $\hat{n}_{\text{eff}}/(\text{draws})$ settles near 0.2, consistent with the theoretical 0.22, and the curve is fairly smooth. For Metropolis with scale 1.5 the ratio settles near 0.06.
- For the slowly mixing sampler (scale 0.2) the ratio is *highest* for short runs and falls as n grows. On short runs the chain has not yet revealed its long-range autocorrelation, and the truncation rule stops the sum at small lags, so $\hat{n}_{\text{eff}}$ is biased upward. A small $\hat{n}_{\text{eff}}$ that looks stable on a short run should not be trusted.
- So $\hat{n}_{\text{eff}}$ is only a reliable guide once the run is long compared with the autocorrelation time.

8 Exercise 7: Analysis of Survey Data

Problem (paraphrased). Section 8.3 of [1] analyses a stratified pre-election survey with a hierarchical model on the stratum probabilities. (a) Do the computations for the simple nonhierarchical model. (b) Use the Metropolis algorithm for the hierarchical model, with the simulations from (a) as the starting distribution, and check against Figure 8.1b.

8.1 Data

The data are the CBS News survey of 1447 adults, divided into 16 strata (4 regions $\times$ 4 density classes I–IV), with the proportions of Bush, Dukakis and other/no-opinion responses in each stratum and each stratum's proportion [2]. Stratum counts were reconstructed from the proportions as $n_j = \text{round}(1447 \times \text{proportion}_j)$, $y_{j1} = \text{round}(n_j \times \text{Bush}_j)$, $y_{j2} = \text{round}(n_j \times \text{Dukakis}_j)$ and $y_{j3} = n_j - y_{j1} - y_{j2}$.

Assumptions to check against Section 8.3.

- The reconstructed counts sum to 1442 rather than 1447 because of rounding.
- The stratum proportion is treated as the stratum's share of the sample, and the population weights N_j/N are taken equal to it.
- The hierarchical model below is my reconstruction (two logistic-normal layers); it may differ in detail from the model in the book.

8.2 Quantity of interest

Let $\theta_j = (\theta_{j1}, \theta_{j2}, \theta_{j3})$ be the probabilities of Bush, Dukakis and other in stratum j . The population proportion supporting Bush among those expressing a preference is

$$\beta = \frac{\sum_j N_j \theta_{j1}}{\sum_j N_j (\theta_{j1} + \theta_{j2})}. \quad (12)$$

Write $\alpha_j = \theta_{j1}/(\theta_{j1} + \theta_{j2})$ and $\rho_j = \theta_{j1} + \theta_{j2}$. The multinomial likelihood factorizes as

$$y_{j1} \mid m_j \sim \text{Bin}(m_j, \alpha_j), \quad m_j \mid n_j \sim \text{Bin}(n_j, \rho_j), \quad m_j = y_{j1} + y_{j2},$$

and $\beta = \sum_j N_j \rho_j \alpha_j / \sum_j N_j \rho_j$.

8.3 (a) Nonhierarchical model

Independent uniform (Dirichlet(1, 1, 1)) priors on each θ_j give independent posteriors

$$\theta_j \mid y \sim \text{Dirichlet}(y_{j1} + 1, y_{j2} + 1, y_{j3} + 1),$$

so $\alpha_j \sim \text{Beta}(y_{j1} + 1, y_{j2} + 1)$ and $\rho_j \sim \text{Beta}(m_j + 2, y_{j3} + 1)$, independently. Simulating 4000 draws of each and computing β gives the histogram in Figure 8(a).

8.4 (b) Hierarchical model fitted by Metropolis

$$\text{logit}(\alpha_j) \sim N(\mu_\alpha, \tau_\alpha^2), \quad \text{logit}(\rho_j) \sim N(\mu_\rho, \tau_\rho^2), \quad p(\mu, \tau) \propto 1, \quad (13)$$

for $j = 1, \dots, 16$, with the binomial likelihoods above. The two layers are independent given the data and are fitted separately.

Algorithm. Single-site random-walk Metropolis: in each iteration update each stratum's logit, then μ , then $\log \tau$ (with the Jacobian term $+\log \tau$ for the uniform prior on τ). The jumping scale for stratum j is $2.4/\sqrt{f_j p_j (1 - p_j)}$, the usual normal-approximation scale.

Starting distribution. Each of the 4 chains starts from a different draw of part (a): $\text{logit}(\alpha_j)$ and $\text{logit}(\rho_j)$ are set to the logits of the (a) draw, and μ and τ to their mean and standard deviation across strata. Each chain has 20,000 iterations, the first half is discarded, and the acceptance rates are about 0.2 for the stratum effects and 0.2–0.4 for μ and $\log \tau$.

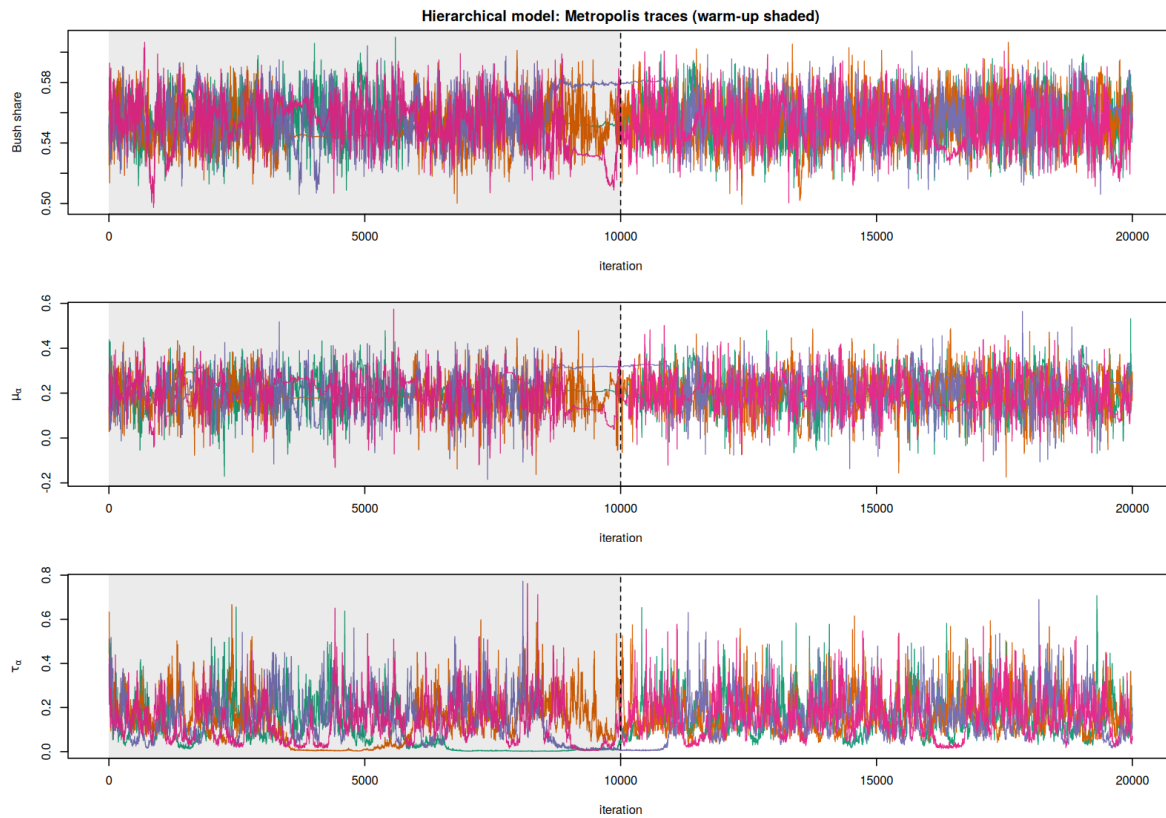


Figure 7: Hierarchical model (centered): Metropolis traces of the population Bush share β , μ_α and τ_α (four chains, warm-up shaded).

Quantity	Mean	SD	$\hat{R}$	ESS
Bush share (population)	0.555	0.014	1.020	224
μ (opinion split)	0.205	0.076	1.012	491
τ (opinion split)	0.161	0.091	1.007	283
μ (response rate)	2.284	0.105	1.043	122
τ (response rate)	0.147	0.119	1.123	29

Table 3: Hierarchical model: posterior summaries, split $\hat{R}$ and $\hat{n}_{\text{eff}}$ from (11.8), computed from $4 \times 10,000$ post-warm-up draws.

8.5 Results

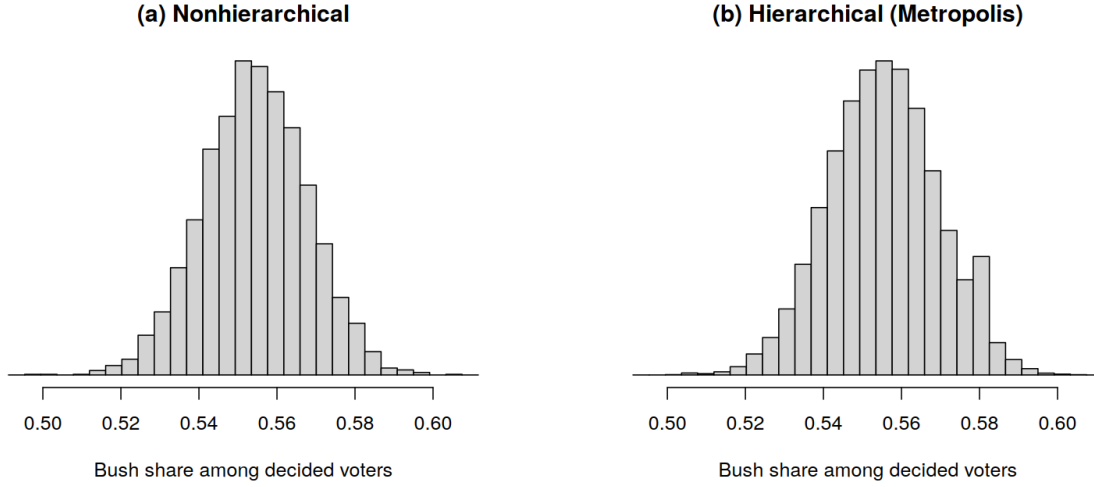


Figure 8: Posterior draws of the Bush share among decided voters: (a) nonhierarchical model, (b) hierarchical model.

Model	Mean	SD	95% interval
(a) Nonhierarchical	0.554	0.013	[0.528, 0.581]
(b) Hierarchical	0.555	0.014	[0.528, 0.582]

Table 4: Bush share among decided voters.

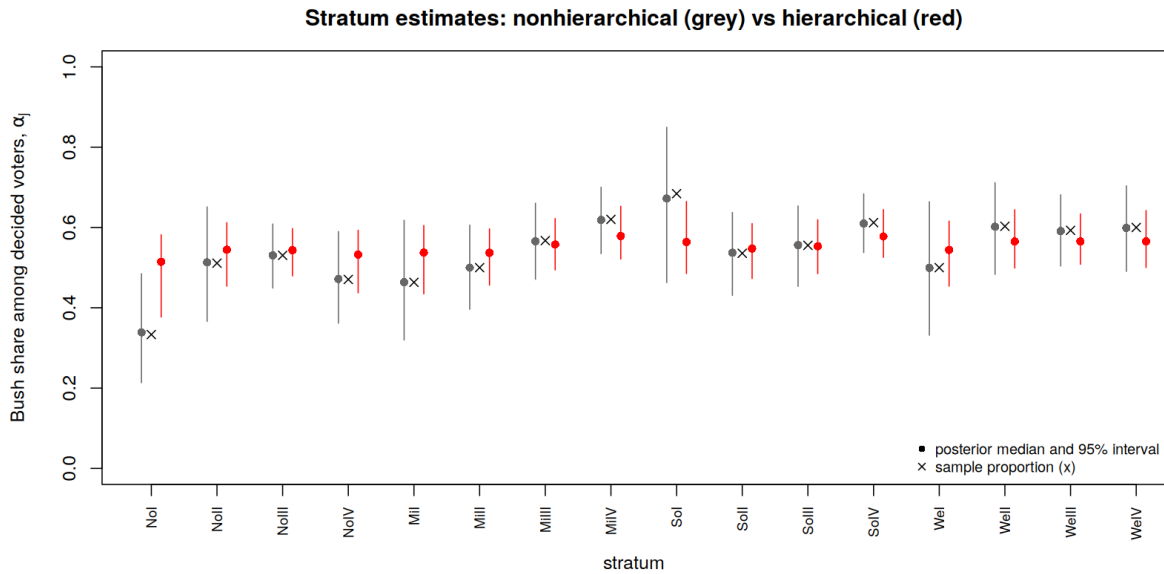


Figure 9: Stratum-level Bush share among decided voters α_j : posterior medians and 95% intervals from the nonhierarchical (grey) and hierarchical (red) models. Crosses are the raw sample proportions.

Reading the results.

- The overall population estimate is essentially unchanged by the hierarchical model (0.554 versus 0.555): the strata are large enough in total that the aggregate is well determined.
- The hierarchical model shrinks the stratum estimates toward the common mean. The effect is largest for the smallest strata: South I (22 respondents) moves from about 0.67 to 0.56 and Northeast I (46 respondents) from about 0.34 to 0.51, and their intervals narrow.
- The centered Metropolis chains show long stuck stretches whenever τ is near 0, which is why $\hat{n}_{\text{eff}}$ for β is only about 224 out of 40,000 draws. This is explained in Section 8.7 and fixed in Section 8.8.
- The comparison with Figure 8.1b is in Section 8.6.

8.6 Comparison with Figure 8.1b

The Figure 8.1b histogram you sent is centred near 0.11, has most of its mass between about 0.06 and 0.16, and extends from about 0.03 to 0.22. That cannot be the Bush share itself (about 0.55). It matches the **margin of Bush over Dukakis among decided voters**, $\beta - (1 - \beta) = 2\beta - 1$.

Margin (Bush – Dukakis)	Mean	SD	95% interval	Pr(> 0)
(a) Nonhierarchical, decided voters	0.109	0.027	[0.056, 0.161]	0.9998
(b) Hierarchical (centered), decided voters	0.111	0.028	[0.056, 0.163]	1.000
(b) Hierarchical (non-centered), decided voters	0.111	0.027	[0.057, 0.165]	1.000
(a) Nonhierarchical, all adults	0.098	0.024	[0.050, 0.144]	0.9998
(b) Hierarchical (non-centered), all adults	0.100	0.025	[0.052, 0.150]	1.000

Table 5: Posterior summaries of the margin. “All adults” puts the no-opinion group in the denominator: $\sum_j N_j \rho_j (2\alpha_j - 1) / \sum_j N_j$.

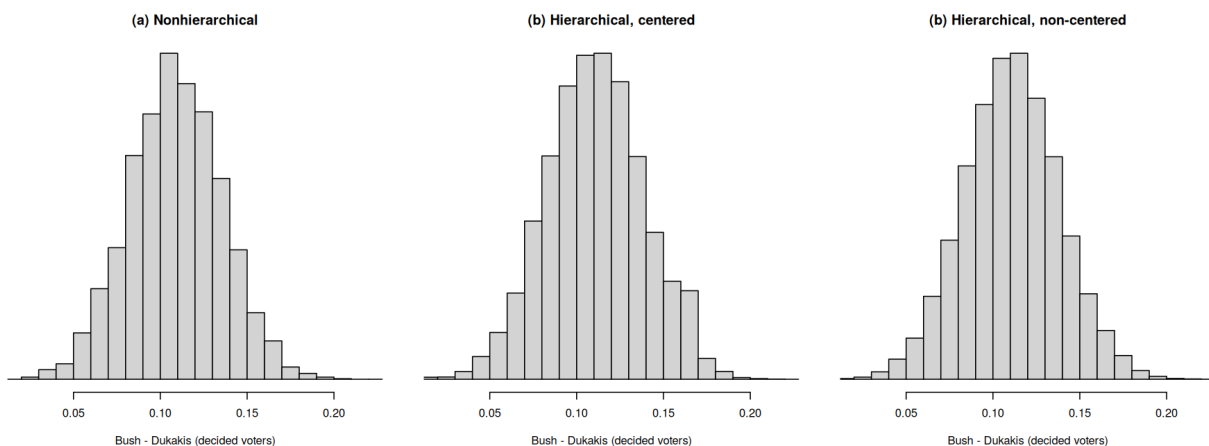


Figure 10: Posterior draws of the margin among decided voters, on the same horizontal range as Figure 8.1b (0.02 to 0.22): (a) nonhierarchical, (b) hierarchical with centered and non-centered samplers.

Reading the comparison.

- The hierarchical histograms are unimodal and roughly symmetric, centred at 0.11, with the bulk between 0.06 and 0.16 and thin tails reaching about 0.03 and 0.20. This agrees with Figure 8.1b by eye.
- The posterior probability that Bush leads is essentially 1 (> 0.9998): the whole distribution lies to the right of zero, and the 95% interval for the lead is about 5.6 to 16.5 percentage points.
- The hierarchical and nonhierarchical margins are almost the same (0.111 versus 0.109); the hierarchical model changes the stratum estimates but barely changes the poststratified aggregate.
- **Caveats.** i inferred the estimand from the axis of the figure using the trace plots. The “all adults” version centres at 0.10 and fits nearly as well on visual assessment.

8.7 Why do some chains look odd? The funnel

In the centered Metropolis traces (Figure 7) two things stand out. In the Bush-share and μ_α panels, one chain (purple, chain 3) sits almost flat from roughly iteration 9000 to 11,000, with shorter flat stretches in other chains. In the τ_α panel, chains drop to almost 0 and stay there for thousands of iterations, while the remaining stretches look like healthy, dense, overlapping noise.

Layer	Chain	Runs	Longest	Iterations in neck	$ \Delta\mu $ inside	$ \Delta\mu $ outside
opinion split	1	1	3387	3387	0.0000	0.012
opinion split	2	1	1746	1746	0.0000	0.014
opinion split	3	2	1783	1965	0.0001	0.013
opinion split	4	1	515	515	0.0001	0.012
response rate	1	2	3718	3918	0.0001	0.013
response rate	2	0	0	0	–	0.016
response rate	3	0	0	0	–	0.016
response rate	4	4	213	741	0.0002	0.012

Table 6: Centered sampler, all 20,000 iterations: runs of at least 100 consecutive iterations with $\tau < 0.03$ (the “neck” of the funnel), and the average absolute change in μ per iteration inside and outside those runs.

What the numbers say. Table 6 shows that four of the four opinion-split chains spend long, unbroken stretches with $\tau < 0.03$: chain 1 for 3387 iterations, chain 2 for 1746, chain 3 for 1783 (and 1965 in total), and chain 4 for 515. Inside these stretches μ moves by 0.0000–0.0001 per iteration, compared with 0.012–0.014 outside, so the chain is effectively frozen. The same happens in the response-rate layer (chain 1: 3718 iterations in the neck).

Why it happens.

1. *It is a property of the posterior, not a bug.* With a uniform prior on τ the posterior has real mass near $\tau = 0$: about 8% of the post-warm-up draws of τ_α are below 0.05. The data are consistent with all strata sharing nearly the same Bush share.
2. *The funnel shape.* When τ is small, every λ_j must lie within about τ of μ ; when τ is large the

λ_j can spread out. The joint posterior of $(\tau, \lambda_j - \mu)$ is therefore a funnel, wide at large τ and a thin neck at small τ (left panel of Figure 11).

3. *Why the sampler gets trapped.* The single-site proposals for λ_j have scale about 0.3–0.5, set by the data. Inside the neck the allowed range for each λ_j is only $\tau < 0.03$, so almost every proposal is rejected. The proposal for μ (scale 0.2) is rejected too, since it would leave all 16 values of λ_j behind. And τ cannot grow unless the λ_j spread out first, because the density of 16 values near μ drops sharply as τ increases. The λ_j cannot spread until τ grows: a chicken-and-egg trap broken only by a lucky acceptance.
4. *Why Bush share is flat at a different level.* In the neck all strata share nearly the same $\alpha_j \approx \text{logit}^{-1}(\mu)$. Chain 3 froze at $\mu_\alpha \approx 0.3$, above the posterior mean 0.205, so its Bush share sat near 0.58 instead of 0.555.

Consequences. Because the stuck stretches differ between chains, $\hat{R}$ is inflated for some quantities ($\hat{R} = 1.12$ for τ_ρ) and the effective sample sizes are small ($\hat{n}_{\text{eff}} = 29$ for τ_ρ and 224 for the Bush share, out of 40,000 draws). Chain 3’s frozen stretch extends past the end of the warm-up half, so discarding half the chain did not remove it. The point estimates are still reasonable (Bush share 0.5554, margin 0.1109) because the stuck segments are a small share of the draws, but their Monte Carlo error is much larger than the number of draws suggests.

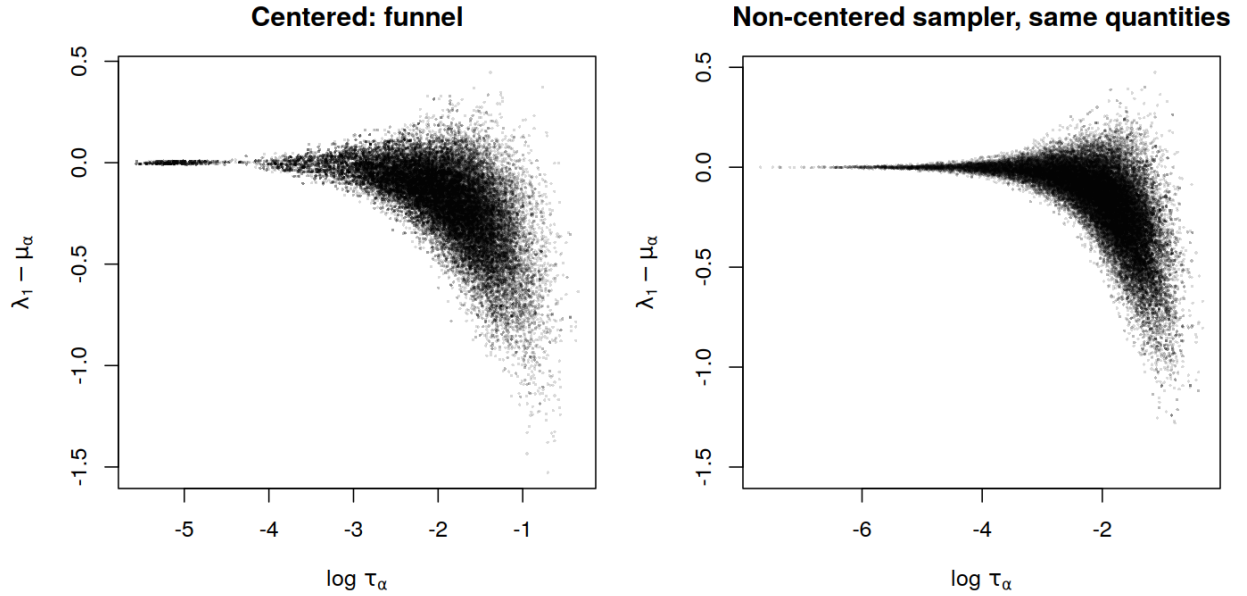


Figure 11: Left: $\lambda_1 - \mu_\alpha$ against $\log \tau_\alpha$ for the centered chains. The thin horizontal cluster near $\log \tau_\alpha \approx -5$ is the neck where chains get stuck. Right: the same quantities from the non-centered sampler, which moves smoothly down the funnel (to $\log \tau_\alpha \approx -7.5$) and back.

8.8 Fix: non-centered parameterization

Write $\lambda_j = \mu + \tau \eta_j$ with $\eta_j \sim N(0, 1)$ a priori, and run Metropolis on $(\eta_1, \dots, \eta_{16}, \mu, \log \tau)$. The prior on each η_j no longer depends on τ , so the funnel disappears: the proposal scale for η_j is $\min(2.4 \times (\text{data scale})/\tau, 2.4)$. Acceptance rates are about 0.40–0.42 for the η_j , 0.41–0.56 for μ and

0.71–0.79 for $\log \tau$ (the last could be increased with a larger jumping scale).

Quantity	Centered		Non-centered	
	$\hat{R}$	ESS	$\hat{R}$	ESS
Bush share (decided)	1.020	224	1.000	9419
Margin (decided)	1.020	224	1.000	9419
μ_α	1.012	491	1.001	3156
τ_α	1.007	283	1.004	1194
μ_ρ	1.043	122	1.001	3429
τ_ρ	1.123	29	1.003	985

Table 7: Centered versus non-centered Metropolis: split $\hat{R}$ and $\hat{n}_{\text{eff}}$ from (11.8), computed from $4 \times 10,000$ post-warm-up draws.

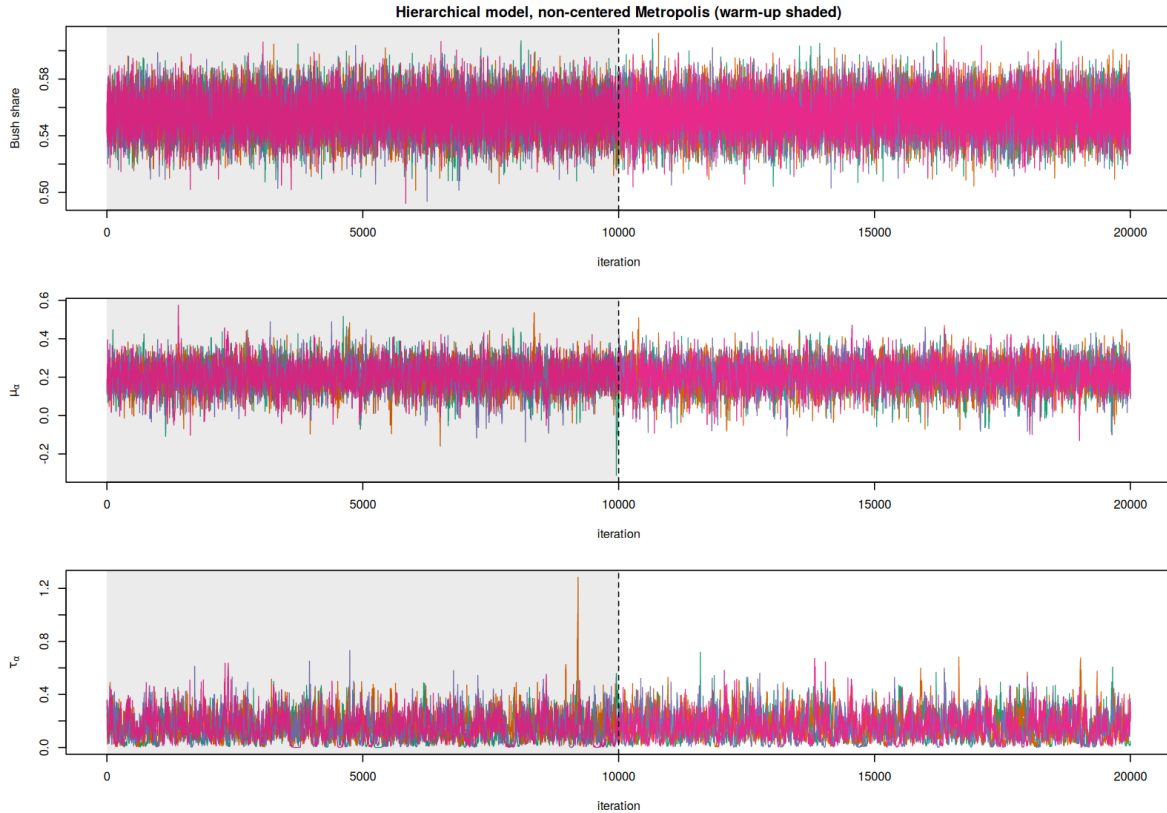


Figure 12: Non-centered sampler: Bush share, μ_α and τ_α show dense overlapping traces with no stuck stretches.

Result. Every $\hat{R}$ is at most 1.004 and the ESS for the Bush share rises from 224 to 9419 ($\approx 42\times$), for τ_ρ from 29 to 985, for τ_α from 283 to 1194 and for μ_α from 491 to 3156. The estimates themselves agree (Bush share 0.5554 versus 0.5553, margin 0.1109 versus 0.1106), which confirms that the centered chains were sampling the right distribution, only very inefficiently.

Other remedies.

- Use a weakly informative prior on τ (for example half-Cauchy) to put less mass in the neck.
- Use Gibbs steps for μ and τ and a Metropolis step for the λ_j ; this helps but does not remove the funnel.
- Run longer chains from more dispersed starts and always check $\hat{R}$ and $\hat{n}_{\text{eff}}$ for the hyperparameters, not only for the quantity of interest.

9 Detailed Interpretation of the Plots

9.1 Bivariate normal example (Section 6)

Gibbs trace plot. The four chains, started from $(\pm 4, \pm 4)$, are indistinguishable from the first few iterations onward. This is expected: each full Gibbs sweep multiplies the deviation of θ_1 from its mean by about $\rho^2 = 0.64$, so the starting point is forgotten geometrically. The traces look like a dense, overlapping band (a “hairy caterpillar”) with no trends or flat stretches, and there is no visible difference between the warm-up half and the kept half. Together with $\hat{R} \approx 1.00$ this is the picture of a converged, well-mixing sampler.

Metropolis, jumping scale 0.2. The chains take roughly 200–300 iterations just to travel from ± 4 to the bulk, and afterwards they wander in slow waves lasting hundreds of iterations. Even after warm-up the chains sit in different places at the same time (one near -1 while another is near $+1$), so the between-chain variance is not negligible: $\hat{R} \approx 1.11$ and 1.08 , above the 1.1 threshold for θ_1 . The acceptance rate of 0.84 is high but misleading: almost every proposal is accepted because the steps are tiny.

Metropolis, jumping scale 1.5. The chains converge within about 50–100 iterations and move much faster, but the trace is visibly stepped. The flat horizontal pieces are rejected proposals (the acceptance rate is 0.27 , so on average a chain stays put for about 3–4 iterations). $\hat{R} \approx 1.01$.

Scatter plots. All three clouds are elongated along the diagonal, as the correlation $\rho = 0.8$ requires, and centred at the origin, which suggests all three are sampling approximately the right distribution. The Gibbs cloud fills the red 95% ellipse smoothly. The scale-0.2 cloud is visibly too narrow, with thin tails (estimated standard deviations 0.78 and 0.81 instead of 1): the chains have simply not had time to explore the tails. The scale-1.5 cloud looks sparse because repeated draws are plotted on top of one another: only roughly a quarter of the 4000 draws are distinct points.

Autocorrelation plots. For Gibbs the lag-1 autocorrelation is about 0.62 , close to the theoretical $\rho^2 = 0.64$, and it is negligible after about 7 lags; the small bumps at lags 20–55 are sampling noise from only 1000 draws of a single chain, not real dependence. For scale 0.2 the autocorrelation starts at about 0.98 and is still about 0.15 at lag 60. For scale 1.5 it starts at about 0.85 and is near zero by lag 40. The integrated autocorrelation time ($4000/\hat{n}_{\text{eff}}$) is therefore about 4 draws for Gibbs, 13 for scale 1.5 and 108 for scale 0.2: one effective draw costs about 4, 13 or 108 iterations respectively.

9.2 Exercise 6: $\hat{n}_{\text{eff}}$ over time

Left panel. $\hat{n}_{\text{eff}}$ grows roughly linearly with run length for the Gibbs sampler (185 at $n = 400$, 4073 at $n = 10,000$) and for Metropolis with scale 1.5 (74 to 1314). For scale 0.2 it creeps from 7 to

only 70, so after $4 \times 5000 = 20,000$ post-warm-up draws there are about 70 effectively independent ones.

Right panel. The efficiency per draw is the slope of the left panel. For Gibbs it is noisy for short runs (0.32 at $n = 200$, dipping to about 0.18 near $n = 800$) and settles at about 0.20–0.23, around the theoretical 0.22 (dotted line). For scale 1.5 it settles near 0.06 and for scale 0.2 near 0.004. The early noise comes from poorly estimated autocorrelations in short chains.

Practical meaning. To obtain 1000 effective draws one needs about 4500 Gibbs draws, about 16,000 draws with scale 1.5 and several hundred thousand with scale 0.2. A chain whose $\hat{n}_{\text{eff}}$ is still climbing, or whose efficiency per draw is still moving, has not run long enough for the estimate to be trusted.

9.3 Exercise 7: survey model

Histograms of the Bush share. Both are symmetric and bell-shaped. The nonhierarchical model gives mean 0.554, sd 0.013 and 95% interval [0.528, 0.581], and the hierarchical model gives 0.555, 0.014 and [0.528, 0.582]. The small difference in spread is within the Monte Carlo error of the centered hierarchical run (its $\hat{n}_{\text{eff}}$ is only 224, which gives a relative error of about 5% on a standard deviation), and the non-centered run gives 0.0137.

Stratum-level (shrinkage) plot. In the nonhierarchical model the posterior medians (grey) sit almost exactly on the sample proportions (crosses), with wide intervals for small strata. In the hierarchical model (red) every stratum is pulled toward the common mean of about 0.55: all red medians lie between about 0.51 and 0.58, whereas the raw proportions range from about 0.34 to 0.69. The pull is strongest where the data are weakest: Northeast I moves from about 0.34 to 0.51 and South I from about 0.67 to 0.56, and their intervals shrink sharply. Large strata hardly move. The size of the shrinkage reflects $\tau_\alpha \approx 0.15$ on the logit scale (about 3.7 percentage points): most of the apparent differences between strata are consistent with sampling noise, so the hierarchical estimates are nearly pooled.

Margin histograms. See Section 8.6: the hierarchical margin is centred at 0.11, matching the shape and location of Figure 8.1b, and the three panels (nonhierarchical, centered, non-centered) are nearly identical.

Centered trace plots, funnel plot and non-centered traces. See Sections 8.7 and 8.8. In short: the flat stretches are the sampler stuck in the neck of a funnel (small τ), which makes the centered chains very inefficient but does not change the answer, and a non-centered parameterization removes the problem.

What a healthy trace looks like. A good trace (like the Gibbs traces, the non-centered traces, or the non-stuck stretches of the centered traces) is dense and stationary with a constant mean and spread, the chains overlap completely, and no chain has long flat or trending segments. Flat stretches mean a chain is not moving; slow drifts mean it has not converged; chains that disagree mean $\hat{R}$ will be large. Always pair the trace plots with $\hat{R}$ and $\hat{n}_{\text{eff}}$ for every parameter, hyperparameters included.

References

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- [3] Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., and Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC. *Bayesian Analysis*, 16(2):667–718.